Adam Danicki he/him/his

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SUMMARY

Software engineer with experience building REST APIs, database-backed systems, AI/document-processing backends, recommendation services, and reusable React components. Strong foundation in Python, JavaScript/TypeScript, SQL, FastAPI, React, PostgreSQL, SQLAlchemy, pgvector, Docker, API design, relational data modeling, embeddings, vector search, LLM integration, debugging, and Git-based development. Built projects involving RAG document search, a 26-endpoint FastAPI/PostgreSQL API, a recommendation API, and a React/TypeScript degree planning platform.

EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences

Bachelor of Science in Computer Science, May 2025 | GPA 3.408 (Spring 2025 term GPA 3.936)

Relevant coursework: Software Engineering, Operating Systems, Computer Networks, Algorithms, Machine Learning, Statistics, Linear Algebra, Computer Systems Principles, Programming with Data Structures, Reasoning Under Uncertainty

PROJECTS

Clinical Document RAG Assistant | Python, FastAPI, PostgreSQL, pgvector, SQLAlchemy, OpenAI API, Docker

- Built a backend RAG system for PDF-based clinical-style documents, supporting file upload, text extraction, overlapping chunking, OpenAI embedding generation, PostgreSQL/pgvector storage, semantic search, and source-grounded answer generation.
- Designed SQLAlchemy models for uploaded documents and document chunks with one-to-many relationships, persistent metadata, chunk content, and vector embeddings so retrieved answers can be traced back to source PDFs.
- Implemented modular FastAPI services and endpoints for document ingestion, semantic search, and question answering, returning source previews through documented OpenAPI/Swagger interfaces.

Longitudinal Lesion Tracking REST API | Python, FastAPI, SQLAlchemy, PostgreSQL, Docker Compose

- Built a Dockerized PostgreSQL-backed REST API with 26 endpoints supporting CRUD operations, nested read views, and analytics-style queries for structured longitudinal measurement records.
- Designed a normalized relational schema with foreign keys, cascading deletes, scoped uniqueness constraints, and validation checks to prevent duplicate or inconsistent records.
- Created FastAPI/OpenAPI documentation and automated demo data workflows to support repeatable testing, endpoint validation, and handoffs.

Song Recommender API | Python, FastAPI, SQLAlchemy, PostgreSQL, scikit-learn, Docker

- Built a catalog-backed recommendation API supporting Search → select up to 20 seed tracks → return up to 20 similarity-based recommendations through stable request/response schemas.
- Created an offline indexing pipeline using StandardScaler, optional PCA embeddings, and cosine NearestNeighbors retrieval, persisting fitted artifacts plus feature-order metadata to keep recommendations reproducible across runs without retraining.
- Focused on consistent outputs and predictable data contracts to support downstream analysis and troubleshooting when inputs drift or data quality issues appear.

CICS Degree Planner | React, TypeScript, HTML/CSS, API, Docker

- Developed React and TypeScript components for a multi-semester degree planning platform with course selection, schedule visualization, and interactive planning workflows.
- Integrated structured course data into frontend state and reusable UI components while maintaining responsive behavior during frequent user interactions and state changes.
- Collaborated on feature design, debugging, code integration, and iterative delivery within a 6-person Agile team using Docker-based development environments.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, HTML/CSS, C, C++
Backend and APIs: FastAPI, Node.js, Express, REST APIs, OpenAPI/Swagger, JSON, OAuth, session-based authentication
Frontend: React, reusable components, state management, responsive UI, drag-and-drop interfaces
Databases: PostgreSQL, SQLAlchemy, relational modeling, joins, aggregations, constraints, validation rules
ML and data: scikit-learn, NumPy, pandas, Matplotlib, embeddings, vector search, LLM integration, RAG pipelines, feature engineering
Tools: Git/GitHub, Linux, Docker, Docker Compose, Agile/Scrum fundamentals

EXPERIENCE

Patient Care Technician - Hematology/Oncology (Boston, MA) | Beth Israel Deaconess Medical Center — April 2026 - Present

- Document patient information in Epic EHR workflows, reviewing vitals for accuracy or concerning changes and communicating findings in a high-acuity hematology/oncology setting.
- Monitor patient symptoms and condition trends during shifts, escalating abnormal findings to nursing staff to support timely care.

IT Support Administrator (Boston, MA) | Pope John Paul II Polish Language School — September 2017 - August 2023

- Provided onsite technical support for instructors and staff by troubleshooting classroom computers, Wi-Fi connectivity, display systems, printers, AV equipment, and setup issues during live weekend school operations.